AI Civilization Simulator: Product Documentation

Project Code Name: Digital Petri Dish
Version: 0.1 (Phase 0 - Architecture Design)
Date: March 2026
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Executive Summary

AI Civilization Simulator is an experimental platform that uses artificial intelligence to create and observe virtual societies at scale. The system simulates thousands of autonomous agents living in a persistent world, where civilizations emerge, evolve, and interact without predetermined outcomes. By combining lightweight rule-based agents with strategic LLM-driven leaders, the simulator creates a computational laboratory for studying emergent social phenomena—from economic cooperation to collective revolts and inter-civilization conflicts.

Unlike traditional simulations with scripted behaviors, this platform allows genuine emergence: individual agents pursue their needs within environmental constraints, leading to unpredictable macro-level patterns. The system serves as both a research tool for computational social science and a creative exploration of how complex societies arise from simple individual rules.

Vision & Objectives

Core Vision

Create a "living petri dish" where AI-driven societies grow, compete, and evolve—enabling observation of emergent behaviors that mirror real-world social dynamics without human-authored scripts.

Primary Objectives

1. **Emergent Complexity:** Design a system where macro-level phenomena (revolutions, wars, trade networks, cultural evolution) emerge naturally from micro-level agent interactions
2. **Scalability:** Support 1000+ simultaneous agents through hybrid intelligence architecture (rule-based + LLM-driven)
3. **Cost Efficiency:** Maintain viable operational costs by strategically deploying expensive LLM calls only where human-like reasoning is essential
4. **Observability:** Provide rich visualization and analytics to track civilization evolution, detect emergent patterns, and generate narrative reports
5. **Interactivity:** Enable "god mode" intervention—inject disasters, modify physics rules, observe butterfly effects

Research Value

The platform enables investigation of questions at the intersection of AI, sociology, and complexity science:

- How do decentralized agents self-organize into hierarchical societies?
- What conditions trigger collective action cascades (revolutions, mass migrations)?
- Can AI societies develop emergent communication protocols or cultural artifacts?
- How do information asymmetries between social layers affect governance stability?
- What role does individual agent diversity play in civilization resilience?

System Overview

High-Level Architecture

The simulator operates on a **three-layer agent hierarchy** embedded within a **rule-based persistent world**. The architecture deliberately separates lightweight computational agents (majority) from

expensive LLM-powered strategic agents (minority), enabling large-scale simulation within practical token budgets.

[Conceptual Architecture Diagram]
Figure 1: Three-tier agent pyramid with bidirectional information flow and environment feedback loops

Architectural Principles

1. **Hierarchical Intelligence:** Intelligence complexity matches social responsibility—bottom-layer agents use finite state machines, top-layer leaders use frontier LLMs
2. **Emergent Causality:** No predetermined storylines; all macro events arise from micro-level agent decisions and environmental feedback
3. **Information Asymmetry:** Higher layers receive aggregated/filtered information, mirroring real-world governance structures
4. **Deterministic Environment:** World physics (resources, geography, economics) follows objective rules, not LLM hallucinations
5. **Asynchronous Communication:** Agents interact via message queues, enabling scalable parallelization

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Agent Architecture

The system implements a **three-tier social pyramid** where agent complexity increases with hierarchical position.

Layer 1: Civilian Agents (Population Base)

Scale: 1000+ agents
Intelligence Model: Finite State Machine + Markov Chain
LLM Dependency: Zero
Computational Cost: Near-zero (pure matrix operations)

Purpose & Design Philosophy

Civilian agents represent the population base—farmers, workers, traders, families. Individually they have minimal influence, but collectively they form the substrate from which macro-social phenomena emerge[80][83]. Their behavior is modeled using probabilistic state machines inspired by NPC AI in game simulations[77][79], but enhanced with social contagion mechanisms.

State Space

Each civilian exists in one of seven discrete states at any time:

State	Description
LABOR	Productive work (farming, crafting, building)
REST	Recovery and consumption of resources
TRADE	Economic exchange with other agents
SOCIALIZE	Non-economic interaction (relationship building)
MIGRATE	Moving to new location seeking better conditions
PROTEST	Active resistance to governance policies
COMBAT	Violent conflict (war, rebellion, self-defense)

Table 1: Civilian agent state definitions

Transition Mechanics

State transitions occur probabilistically each simulation tick, governed by a **dynamic transition matrix**:

$$P_{ij}(t) = f(\mathbf{M}_{base}, \mathbf{E}(t), \mathbf{N}(t), \mathbf{A})$$

Where:

- $\mathbf{M}_{base}$ = Agent's personality-based baseline transition matrix
- $\mathbf{E}(t)$ = Environmental variables (resource scarcity, tax rate, violence level)
- $\mathbf{N}(t)$ = Neighborhood state distribution (social contagion)

- $\mathbf{A}$ = Agent's intrinsic attributes (age, occupation, tolerance threshold)

Granovetter Threshold Model

The most critical mechanism for emergent collective action is the **threshold-based protest contagion**[81][84]. Each agent has a personal threshold $\theta_i \in [0, 1]$ representing the proportion of neighbors that must be protesting before the agent joins:

$$\text{Protest Probability} = \begin{cases} 0.05 & \text{if } r < \theta_i \\ 0.95 & \text{if } r \geq \theta_i \end{cases}$$

where r = ratio of protesting neighbors.

This creates **tipping point dynamics**—grievances can simmer quietly until a critical threshold is crossed, triggering cascading mass protests[83][89]. The threshold distribution across the population determines whether a society experiences:

- Successful revolution (broad participation)
- Sustained civil unrest (moderate participation)
- Effective repression (insufficient participation)

Attributes

Each civilian agent possesses persistent attributes:

Attribute	Type	Effect
Personality	Enum	Affects baseline transition probabilities
Occupation	Enum	Determines resource generation and vulnerability
Age	Integer	Modifies risk tolerance and energy levels
Wealth	Float	Influences trade behavior and migration decisions
Loyalty	Float [0-1]	Resistance to protest despite peer pressure
Threshold	Float [0-1]	Granovetter protest activation threshold
Relationships	Graph	Social network connections (family, friends)

Table 2: Civilian agent attribute schema

Layer 2: Governor Agents (Middle Management)

Scale: 20-50 agents
Intelligence Model: Lightweight LLM (Claude Haiku/Sonnet tier)
LLM Call Frequency: Once per N simulation ticks (e.g., "quarterly")
Computational Cost: Moderate and controlled

Purpose & Design Philosophy

Governors represent regional administrators—mayors, prefects, district managers. They bridge the gap between strategic leadership (top layer) and population reality (bottom layer). Critically, governors operate under **information asymmetry**—they see aggregated statistics but not individual civilian states[86], mirroring real-world bureaucratic limitations.

Inputs & Observability

Each governor receives a periodic intelligence report containing:

- **Population Statistics:** Total count, age distribution, occupation breakdown

- **Economic Metrics:** Average wealth, resource production, trade volume
- **Sentiment Indicators:** Protest rate, loyalty distribution, migration rate
- **Environmental Data:** Resource stockpiles, infrastructure status
- **Superior Orders:** Directives from top-layer leaders

Governors *cannot* see individual civilian decisions or social network dynamics—they must infer population mood from aggregate signals.

Decision Space

Governors make regional policy decisions within constraints:

Decision Type	Options
Tax Rate	Adjust local taxation (within bounds set by leaders)
Resource Allocation	Distribute resources between infrastructure, welfare, military
Security Policy	Increase/decrease policing, employ repression tactics
Economic Incentives	Subsidies, trade regulations, labor policies
Communication	Request aid from leaders, coordinate with neighboring governors

Table 3: Governor decision domains

LLM Prompt Structure

Governor prompts are concise and structured to minimize token consumption:

\begin{verbatim}
You are Governor of Region X in Year Y.
SITUATION:
- Population: 2,340 (15% protesting, up from 3% last quarter)
- Resources: 80% of target stockpile
- Economy: Trade down 22%, average wealth declining
- Orders: Central government demands 10% tax increase

CONSTRAINTS:

- Tax range: 8-15% (currently 10%)
- Security budget: 500 units available

DECISION: What actions do you take? Format as JSON:
{"tax": X, "security": Y, "resource_allocation": {...},
"message_to_leader": "..."}
\end{verbatim}
The lightweight LLM generates a decision object which the simulation engine executes, updating environmental variables that civilian agents react to.

Layer 3: Leader Agents (Strategic Command)

Scale: 3-8 agents
Intelligence Model: Frontier LLM (Claude Opus/GPT-4 tier)
LLM Call Frequency: Major cycles + event-driven triggers
Computational Cost: Primary expense, but low volume

Purpose & Design Philosophy

Leader agents represent civilization heads—kings, presidents, tribal chiefs. They possess the most sophisticated cognitive capabilities: long-term memory, strategic reasoning, natural language diplomacy, and ideological vision. Leaders operate at the slowest timescale but make the highest-impact decisions[60].

Cognitive Architecture

Unlike civilians and governors, leaders have rich internal models:

- **Episodic Memory:** Full history of diplomatic interactions, treaties, betrayals
- **Strategic Knowledge:** Understanding of military capabilities, technological advantages, economic dependencies
- **Personality & Ideology:** Persistent traits affecting negotiation style and policy preferences
- **Theory of Mind:** Models of other leaders' likely intentions and responses

This architecture mirrors AgentSociety's "Mind-Behavior Coupling" approach[36], where internal cognitive state drives external actions.

Decision Domains

Leaders control civilization-level strategy:

Domain	Examples
Foreign Policy	Declare war, propose alliances, negotiate treaties, economic sanctions
Technology	Choose research directions (agriculture, military, infrastructure)
Governance	Set tax policy boundaries, legal frameworks, propaganda campaigns
Resource Strategy	Allocate national reserves, trade agreements, emergency responses
Ideology	Promote cultural values, religious policies, education focus

Table 4: Leader strategic decision domains

Natural Language Diplomacy

The most distinctive feature of leader agents is **multi-turn diplomatic negotiation**. When leaders communicate, they use full natural language:

\begin{verbatim}
Leader A (Kingdom of Westholm) to Leader B (Coastal Federation):
"We propose a mutual defense pact: if either nation faces external aggression,
the other commits 30% of military forces to joint defense. In exchange, we
offer access to our iron mines at 15% below market rate. This benefits us
both given the growing threat from the Northern Tribes."
[Leader B's LLM generates response considering: relationship history, current
military situation, economic needs, trust level, domestic pressures...]
\end{verbatim}

This creates genuine emergent geopolitics—alliances, betrayals, and conflicts arise from agent reasoning, not scripted events.

Environmental Systems

The world provides objective, rule-based structure within which agents operate. Environmental systems are **deterministic**—avoiding the unreliability of LLM-generated environments[36][48].

Spatial Structure

- **Grid-based 2D map:** Each cell has terrain type (plains, forest, mountain, water), resource deposits, and climate
- **Distance mechanics:** Movement costs, trade efficiency, and communication latency based on actual pathfinding
- **Territory control:** Regions belong to civilizations, borders can shift through conquest or negotiation

Resource Economy

Three resource types form the economic foundation:

Resource	Source	Purpose
Food	Farming (civilians in LABOR state)	Necessary for population survival
Materials	Mining, forestry	Used for infrastructure and military
Wealth	Trade surplus	Enables technology research and luxury

Table 5: Core resource types and functions

Resource scarcity directly modifies civilian transition matrices—when food drops below threshold, migration and protest probabilities surge.

Social Networks

Agent relationships form dynamic graphs:

- **Family ties:** Strong bonds, high trust, prioritized in cooperation
- **Friendships:** Moderate bonds, influence social contagion
- **Economic partnerships:** Trade relationships, alliance networks

Social networks determine information flow and collective action coordination—agents with high centrality can become emergent "movement leaders" even at civilian level.

Emergent Phenomena

The system is designed to produce macro-level patterns not explicitly programmed:

Revolution Cascades

When government policies (high taxes, resource shortages) push civilian discontent beyond critical thresholds, Granovetter dynamics can trigger sudden mass uprisings[81][83]. The cascade progresses:

1. Initial unrest: Small percentage of low-threshold agents enter PROTEST state
2. Social contagion: Neighboring agents see protest ratio exceed their thresholds, join
3. Critical mass: Protest rate accelerates exponentially until governors cannot suppress
4. Regime response: Leaders must choose between concessions or violent repression
5. Outcome uncertainty: Revolution may succeed (leader replaced), fail (mass casualties), or lead to civil war

Inter-Civilization Dynamics

Multiple civilizations in the same world create emergent geopolitics:

- **Alliance formation:** Smaller nations ally against powerful hegemons
- **Economic interdependence:** Trade relationships create peace incentives
- **Arms races:** Military technology competition drains resources
- **Proxy conflicts:** Leaders support rebellions in rival territories
- **Cultural spread:** Ideologies propagate through migration and diplomacy

Economic Cycles

Without central planning, market dynamics emerge:

- Resource booms attract migration, leading to overcrowding and eventual bust
- Trade route establishment creates economic integration
- Wealth inequality affects protest susceptibility
- Technology investments change production curves

Information Flow & Asymmetry

A core design principle is that **information degrades and aggregates as it moves up the hierarchy**, while **directives propagate downward but may be distorted by implementation**[80][86].

Upward Flow (Reality → Leaders)

Civilian individual actions
↓ [Aggregation: counts, averages, rates]
Governor regional statistics
↓ [Filtering: governors choose what to report]
Leader strategic intelligence reports
Leaders make decisions based on incomplete, delayed information—they may order war mobilization unaware that their own population is on the brink of revolt.

Downward Flow (Leaders → Reality)

Leader policy directives
↓ [Interpretation: governors adjust to local context]
Governor implementation decisions
↓ [Constraint modification: environmental variables change]
Civilian transition probabilities shift
A leader's "reduce taxes to 12%" becomes a governor's choice of whether to actually comply or report false compliance while maintaining 14% to fund local security—information asymmetry cuts both ways.

Technical Implementation Strategy

Development Phases

Phase	Timeline	Deliverables
Phase 0	2-3 days	Architecture design, technical research documentation
Phase 1	1 week	Minimal world engine: 2D grid + 100 FSM civilians + resources
Phase 2	1 week	LLM integration: 2-3 governor agents with decision loop
Phase 3	1-2 weeks	Emergent layer: trade, alliances, conflict, first cascade observations
Phase 4	2 weeks	Scale-up: 1000+ agents, hybrid intelligence distribution, visualization
Phase 5	1-2 weeks	God mode: intervention tools, parameter tuning, event injection

Table 6: Development roadmap

Tool Allocation

The project leverages a three-tool development stack:

- **Claude Code Max:** Core simulation engine development, agent framework implementation, local testing and iteration
- **Perplexity Max:** Asynchronous cloud execution (overnight simulation runs), technical research and architecture consultation, automated report generation (PDF/PPT outputs)
- **OpenClaw:** Not used in simulator development; reserved for personal productivity automation

Token Cost Optimization

Projected token consumption and mitigation strategies[37][51][56]:

Strategy	Savings	Implementation
Prompt Caching	85-90%	Anthropic cache API for shared world context
Model Cascading	50-65%	Route simple decisions to Haiku, complex to Opus
Behavior Caching	70-90%	Cache and reuse civilian policy decisions
Periodic Calls	80-95%	Governors called every N ticks, not every tick
FSM Bottom Layer	100%	Zero LLM calls for civilian agents

Table 7: Token cost optimization techniques

Expected operational cost: **$5-15 per 1000-tick simulation** with 1000 agents (3 leaders, 30 governors, 967 civilians) using optimized architecture.

Technology Stack

Component	Technology
World Engine	Python 3.11+
Parallel Execution	Ray distributed framework
Agent Communication	MQTT message queue
State Persistence	SQLite (agent memory) + Redis (world state cache)
LLM Gateway	LiteLLM (unified API, model routing)
Visualization	React + D3.js force-directed graphs
Analytics	Pandas + NumPy (pattern detection)

Table 8: Technical stack components

This stack is validated by AgentSociety's 30,000-agent demonstrations[48][55] and proven to support real-time simulation execution.

Use Cases & Applications

Computational Social Science

The platform enables controlled experiments impossible in the real world:

- Test how information flow speed affects revolution success rates
- Measure effects of wealth inequality on civilization stability
- Explore optimal governance structures for different resource environments
- Study how AI agents develop coordination mechanisms without explicit programming

AI Research

The simulator serves as a testbed for multi-agent AI questions:

- Can LLM agents develop emergent communication protocols?
- How do societies evolve when different agent tiers use different model architectures?
- What role does agent memory architecture play in diplomatic stability?
- Can civilizations "discover" game-theoretic concepts like Nash equilibria through experience?

Creative Exploration

The system generates emergent narratives:

- Live-stream civilization evolution with audience voting on disaster events
- Generate historical narrative reports of AI-civilization rise and fall
- Create "alternate history" scenarios by changing initial conditions
- Produce documentary-style analysis of specific emergent events (e.g., "The Great Westholm Revolt of Year 47")

Educational Applications

The platform demonstrates complex systems concepts:

- Visualize how micro-level rules produce macro-level patterns
- Teach agent-based modeling and emergence theory
- Illustrate game theory, collective action problems, and information asymmetry
- Provide interactive laboratory for sociology and political science students

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Success Metrics

Technical Metrics

- **Scale:** Successfully simulate 1000+ agents in real-time or faster
- **Stability:** Run continuous simulations for 1000+ ticks without crashes
- **Cost:** Maintain operational cost under $20 per 1000-tick simulation
- **Emergence:** Demonstrate at least 3 unprogrammed macro phenomena (revolution, alliance formation, economic cycles)

Research Metrics

- **Replication:** Reproduce known social science findings (Granovetter thresholds, collective action cascades)
- **Novelty:** Identify at least 2 emergent patterns not predicted in advance
- **Publication:** Produce conference-quality paper on findings
- **Open Source:** Release framework for research community use

Creative Metrics

- **Narrative Richness:** Generate coherent "historical accounts" of simulated events
- **Unpredictability:** Same initial conditions produce diverse outcomes across runs
- **Engagement:** System produces genuinely surprising and interesting behaviors

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Risk Assessment

Technical Risks

Risk	Likelihood	Mitigation
Token cost explosion	Medium	Strict prompt caching, FSM bottom layer, periodic calls
Simulation instability	Medium	Extensive unit testing, incremental complexity
LLM hallucination affecting logic	Low	Rule-based environment, structured outputs
Scalability bottlenecks	Low	Ray framework proven to 30K agents

Research Risks

Risk	Likelihood	Mitigation
No emergence observed	Low	Architecture directly based on validated models
Trivial/boring behaviors	Medium	Iterative tuning, diverse initial conditions
Unable to distinguish emergence from randomness	Medium	Statistical analysis tools, control experiments

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Next Steps

Immediate Actions (Phase 0 Completion)

1. Finalize civilian agent state machine specification
2. Define complete environmental variable schema
3. Select and configure technology stack components
4. Set up development environment with Claude Code Max integration

Phase 1 Entry (Week 1)

1. Implement minimal world engine (2D grid + resource system)
2. Create 100 FSM civilian agents with Markov transition logic
3. Build visualization for agent state distribution
4. Run first emergence test: observe protest cascade formation

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Conclusion

AI Civilization Simulator represents a convergence of agent-based modeling, large language models, and complexity science. By strategically combining cheap rule-based agents with expensive LLM-driven strategists in a hierarchical architecture, the system aims to create a computationally feasible platform for observing emergent social phenomena at scale.

The project's core insight—that civilization-level complexity can arise from simple individual rules operating under information asymmetry within structured environments—offers both research value and creative potential. Through careful architectural design informed by AgentSociety[36][48], Affordable Generative Agents[37], and threshold-based collective behavior models[81][89], the simulator aims to achieve genuine emergence within practical resource constraints.

Success will be measured not by predetermined outcomes, but by the system's ability to surprise its creator—to generate social dynamics, conflicts, and resolutions that were never explicitly programmed but arise naturally from the interplay of thousands of autonomous agents pursuing their individual needs in a shared world.

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